TT-P-52D
AMENDMENT-2
June 23, 1977
SUPERSEDING
Amendment-1
November 25, 1974

FEDERAL SPECIFICATION

PAINT, OIL (ALKYD-OIL) WOOD SHAKES AND ROUGH SIDING

This amendment, which forms a part of Federal Specification TT-P-52D, dated March 6, 1973, was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

PAGE 1

Under Federal Specification: line 1, change "TT-L-11555" to "TT-L-1155".
line 2, change "TT-P-143" to "PPP-P-1892".

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Under American Society for Testing and Materials (ASTM) Standards, delete the following:

D 2038 - Determination of Low Concentration of Lead in Paint.
D 2206 - Determination Low Quantities of Mercury in Paint.

Add the following:

D 562 - Test for Consistency of Paints Using the Stormer Viscosimeter.
D 1210 - Test for Fineness of Dispersion of Pigments in Vehicle System.
D 1296 - Test for Odor of Volatile Solvents and Diluents.
D 1308 - Test for Effect of Household Chemicals on Clear and Pigmented Organic Finishes.
D 1475 - Test for Density of Paint, Varnish, Lacquer and Related Products.
D 2369 - Test for Volatile Content of Paints.
D 2698 - Determination of Pigment Content of Solvent-Type Paints by High Speed Centrifuging.
D 3273 - Resistance to Growth of Mold on the Surface of Interior Coatings in an Environmental Chamber.
E 97 - Daylight Directional Reflectance.

Paragraph 3.1, line 3. Change "0.5" to "0.06".

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Paragraph 3.2.9. Delete in its entirety and substitute: "3.2.9 Resistance to fungus growth. When tested as in 4.3.10, the paint shall show no fungus growth.

Table I. Change Extender maximum requirement of types II and III to "50" and add "65" for type IV. Delete "Mercury content and all requirements".

Table I. Change lead content maximum requirement for all types to "0.06".

Table II, line 1. Change "4.7" to "47.5".

Paragraph 4.3, line 1. Delete "Fed. Test Method Std. No. 141 as" and substitute "the methods".

FSC 8010

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Table III. Delete in its entirety and substitute:

TABLE III. Index

Characteristics	Requirement Reference	Applicable Methods: Fed. Test Method Std. No. 141	Applicable Methods: ASTM Method	Para. Ref.
Color	3.1.3	4250	----	--
Condition in container	3.2.1	3011	---	--
Storage stability	3.2.2	----	----	4.3.2
Dilution stability	3.2.3	----	----	4.3.3
Odor	3.2.4	----	----	4.3.4
Application properties	3.2.5	----	----	4.3.5
Flexibility	3.2.6	----	----	4.3.6
Knife test	3.2.7	----	----	4.3.7
Water resistance	3.2.8	----	----	4.3.8
Gloss	Table I	6101	----	4.3.9
Reflectance	Table I	----	E 97	4.3.9
Weight per gallon	Table I	----	D 1475	--
Pigment	Table I	----	D 2698	--
Vehicle solids	Table I	----	D 2698	--
Total solids	Table I	----	D 2369	--
Absorption	Table I	4421	----	--
Fungus resistance	3.2.9	----	D 3273	4.3.10
Consistency	Table I	----	D 562	--
Drying time	Table I	4061	----	--
Coarse particles	Table I	4092	----	--
Fineness of grind	Table I	----	D 1216	--
Hiding power (Contrast ratio)	Table I	4122	----	--
Phthalic anhydride	Table I	7014	----	--
Titanium dioxide (TiO_2)	Table I	7081	----	--
Zinc Oxide (ZnO)	Table I	7331	----	--
Extenders	Table I	7331	----	--
Lead content	Table I	----	----	4.3.11

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Paragraph 4.3.4, line 1. Delete "Method 4401 of Fed. Test Method Std. No. 141" and substitute "ASTM Method D 1296".

Paragraph 4.3.8, line 4. Delete "Method 6011 of Fed. Test Method Std. No. 141" and substitute "ASTM Method D 1308, Section 5d".

Paragraph 4.3.9 Gloss. Add the following: Determine 60° gloss in accordance with Method 6101 of Fed. Test Method Std. No. 141 for compliance with Table I.

Paragraph 4.3.10. Delete in its entirety and substitute: .

4.3.10 Resistance to fungus growth. Determine fungus growth in accordance with ASTM 3273 for compliance with 3.2.9.

Paragraph 4.3.11, and 4.3.11.1. Delete and substitute:

4.3.11 Lead content.

4.3.11.1 Sample preparation. Using a 0.006-inch film applicator and a mechanical applicator plate, duplicate drawdowns for each sample of well-mixed paint shall be made on a standard paint penetration chart and dried for 24 hours. The drawdown shall be at least 10 inches long on the sealed portion of the penetration chart. The drawdown shall be cut into discs of appropriate size to fit the sample holder of a fluorescence X-ray spectrometer.

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4.3.11.2 Procedure. Lead content shall be determined using an X-ray fluorescence spectrometer capable of determining lead content at a minimum level of 0.03 percent by weight of the total nonvolatile. The settings for a wavelength, dispersive fluorescence spectrometer shall be as follows:(1)

Element	Analytical Line	Angle	Crystal	Detection	Collimeter	X-ray tube (MO)
Pb	L	33.93	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Pb (backgrd I)		33.00	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Pb (backgrd II)		35.50	LiF(200)	Flow S.C.	Fine	60Kv 45Ma
Mo	K	20.33	LiF(200)	Flow S.C.	Fine	60Kv 45Ma

Pulse height selection shall be used in all measurements and counting time shall be 100 seconds. Place the sample disc in the wavelength dispersive unit. Measure the count rates of lead, lead background and the Molybdenum Compton scattered background from the X-ray tube.

4.3.11.3 Calculation.

$$R = \frac{I_{Pb} - \frac{I_{Pb\ (Background\ I)} + I_{Pb\ (Background\ II)}}{2}}{I_{Mo}}$$

where I equals gross intensity. These results shall be compared with those obtained by a 0.06 percent lead standard made up from the same type of paint sample and evaluated for compliance with 3.1.

Paragraph 5.1, line 2. Change "TT-P-143" to "PPP-P-1892".
Paragraph 5.5.1, line 1. Change "TT-P-143" to PPP-P-1892".

(1) Energy dispersive fluorescence spectrometers shall be set up according to the manufacturer's manual.

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